

1. List all 6 phases of SDLC in order with one line about each.

1. **Requirement Analysis:** Gathering and defining what the system needs to do based on client or business needs.
2. **Design:** Planning the software architecture, user interface, and system requirements.
3. **Development (Coding):** Writing the actual source code to build the software application.
4. **Testing:** Evaluating the software to find bugs and ensure it meets the initial requirements.
5. **Deployment:** Releasing the finished, tested software to the production environment for users to access.
6. **Maintenance:** Providing ongoing support, updates, and bug fixes after the software goes live.

2. Write 3 differences between Agile and Waterfall models.

- **Flexibility:** Agile is highly adaptable and welcomes changes even late in the process, whereas Waterfall is a strict, sequential process where it's difficult to go back and make changes once a phase is finished.
- **Delivery:** Agile delivers the software in small, usable increments continuously over time, while Waterfall delivers the entire final product all at once at the very end of the project.
- **Customer Involvement:** Agile requires continuous feedback and involvement from the customer throughout development, whereas Waterfall typically only involves the customer heavily during the initial requirement gathering and the final delivery.

3. Name the 3 main Scrum roles and write one line about the responsibility of each.

- **Product Owner:** Acts as the voice of the customer and is responsible for managing the product backlog and prioritizing features to maximize value.
- **Scrum Master:** A facilitator who ensures the team follows Agile/Scrum principles and helps remove any roadblocks slowing the team down.
- **Development Team:** A self-organizing, cross-functional group of professionals who actually design, build, and test the software.

4. What is a Sprint in Scrum? Mention its typical duration.

A Sprint is a short, fixed-length timebox during which a Scrum team works to complete a specific, planned amount of work, resulting in a potentially shippable product increment.

- **Typical duration:** Usually between 1 to 4 weeks, with 2 weeks being the most common.
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5. What do these Jira board columns mean: To Do, In Progress, Done, Backlog?

- **Backlog:** A prioritized master list of all features, ideas, and bugs that need to be worked on eventually, but aren't in the current work cycle.
- **To Do:** Tasks that have been selected from the backlog for the current Sprint and are ready to be picked up by a developer.
- **In Progress:** Tasks that team members are actively working on right now.
- **Done:** Tasks that have been fully completed, tested, and meet the team's criteria for completion.